

Package guide

How px4_rta_mm_gpr is laid out, how its parts connect, ahead-of-time JAX compilation, the threads, processes and locks, and every import

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Table of contents

1	Start here	2
2	The repository	3
3	What happens when you run it	4
4	The parts and how they connect	4
4.1	One control tick (100 Hz, control group)	6
4.2	One plan (rollout group, and the worker process)	6
4.3	Wind and gains (state and estimation groups)	6
5	Ahead-of-time JAX compilation	7
5.1	What JAX does with a function	7
5.2	Why the node compiles ahead of time	7
5.3	What is compiled	7
5.4	Configuration and the disk cache	8
5.5	Practical rules	8
6	Concurrency	9
6.1	Threads, the GIL and XLA	9
6.2	Executors	9
6.3	Callback groups	9
6.4	Where the rollout runs (<code>--rollout-backend</code>)	10
6.5	Garbage collection	10
6.6	Timing statistics	10
7	Shared state and locks	11
7.1	The pattern: build, then publish with one assignment	11
7.2	Every shared variable	11
7.3	The locks there are, and why	12
8	The control law, the model and the GP (in brief)	12

9	Logging, analysis and simulation reuse the same parts	13
10	Every import, and what it is for	13
10.1	Standard library	13
10.2	ROS 2	14
10.3	Numerics and control	15
10.4	Data, plots and video	16
10.5	This package's own modules (who imports what)	16
11	On cpp-version: the C++ fast loop	17
12	Where to change what	19
13	Glossary	20

1 Start here

This guide explains the package as it is now, top to bottom. It is the one to read first; the other documents go deep on one topic each:

document	topic
00 (this one)	layout, data flow, AOT compilation, concurrency and locking, imports
01	concurrency in ROS 2 / Python from first principles (executors, GIL, processes, GC)
02	the multithreaded node: options, benchmarks, flight recorder
03	AOT compilation, early-exit rollouts, wind estimation, what caused the crashes
04	GP feedforward, ground floor, LAND backup, the spatial certificate
06	five model fixes (feedback sign, Jacobian domain, row lookup, body-frame velocities, 100 Hz state) and the TV-GPR / GPR × embedding-system comparison
05 (cpp-version)	the C++ fast loop

Code references are `path:line` relative to `packages/px4_rta_mm_gpr/` unless stated otherwise.

2 The repository

```
PX4_RTA_MM_GPR/
  README.md
  docs/                .qmd sources, PDFs in _output/, figure scripts
  tools/run_sitl.sh    PX4 SITL with this project's parameters
  packages/           colcon --base-paths PX4_RTA_MM_GPR/packages
    flight_recorder/  submodule: HDF5 logging (Python + C++ header)
    px4_rta_mm_gpr/   the ROS 2 package (ament_python)
      package.xml, setup.py, setup.cfg, resource/
      launch/         sim_rta_launch.py, only_full_state_relay_launch.py
      scripts/
        px4_rta_mm_gpr.py  entry point: options, executor, shutdown
        rta_mm_gpr_node.py OffboardControl: callbacks, shared state
        data_analysis/    notebooks 01-04, figures/, log_files/
    px4_rta_mm_gpr/    the library (no ROS, except px4_functions)
      jax_mm_rta/       planar model, TVGPR/GPR, rollout, certificate
      jax_nr/          Newton-Raphson tracker (pitch, yaw rates)
      control_kernels.py AOT: control_step, wind_model, linearize
      concurrency/     RolloutPlan, RolloutEngine, backends, LoopStats
      flight_log/       the node's recorder, the log reader
      px4_functions/    arm, land, heartbeat, setpoints
      utilities/        jax_setup, vehicle constants, yaw unwrapping
      sim/              numerical closed-loop simulation
      analysis.py       summaries, plots, tube videos
      test/             pytest: sim + analysis, certificate, lint
```

Why packages/. colcon finds a package by its `package.xml`, and it does not look inside another package's directory. With `flight_recorder` (and, on `cpp-version`, two C++ packages) living in this repository, each package needs its own directory next to the others; `packages/` holds them.

How it builds. `px4_rta_mm_gpr` is an `ament_python` package: colcon runs `setup.py`, which installs the library (`find_packages` finds `px4_rta_mm_gpr` and `scripts`), `package.xml` and the launch files, and creates two console scripts:

command	runs
<code>ros2 run px4_rta_mm_gpr px4_rta_mm_gpr</code>	<code>scripts/px4_rta_mm_gpr.py:main</code> (the node)
<code>ros2 run px4_rta_mm_gpr flight_log</code>	<code>px4_rta_mm_gpr/flight_log/reader.py:main</code> (summary / CSV export of a log)

`flight_recorder` is an `ament_cmake` package (it installs a Python module *and* a C++ header), so colcon `build` builds both from one command. With `--symlink-install` the installed Python files are symlinks to the sources: edits take effect without rebuilding.

Outside dependencies. `px4_msgs` (PX4's ROS 2 messages, in `~/ws_sensor_combined`), and pip packages: `jax`, `immrax`, `control`, `scipy`, `h5py`, `pandas`, `matplotlib`, `imageio-ffmpeg`.

3 What happens when you run it

`ros2 launch px4_rta_mm_gpr sim_rta_launch.py` runs `px4_rta_mm_gpr --sim --log-file log.log`. Then:

1. `main()` (`scripts/px4_rta_mm_gpr.py`) parses the options into a frozen `RuntimeOptions` data-class and computes the log path, `<workspace>/src/data_analysis/log_files/px4_rta_mm_gpr/`.
2. `rclpy.init()`, then `OffboardControl(sim, options)` (`scripts/rta_mm_gpr_node.py:190`). The constructor does all the slow work *before* anything flies:
 - a. vehicle constants (`utilities/sim_utilities.py` or `hardware_utilities.py`), wind EKF, callback groups, publishers and subscriptions;
 - b. `init_jit_compile_nr_rta()`: tunables, GP buffers, `RolloutConfig`, then **compilation**: the rollout worker process is spawned first (it compiles the rollout in parallel), then the control kernels are compiled ahead of time, the first LQR gains computed, and the node waits for the worker’s “ready” (about 4.4 s in total with a warm disk cache);
 - c. the `FlightRecorder` is created with every option as metadata;
 - d. `gc.collect()`; `gc.freeze()` and the generation-2 threshold is raised (see Concurrency);
 - e. the **timers are created last**, so no callback runs before the constructor finishes.
3. The **executor** is built (`--executor multi|single|events`) and `executor.spin()` runs the callbacks. With autosave on, the recorder flushes to disk every 5 s from its own thread.
4. The **mission**, on the node’s clock `t = now() - T0`:

time	who flies	what the node does
$t < 15$ s	PX4 position controller	position setpoint (0, 4, -12.5) m in SITL; heartbeat; after 10 heartbeats: offboard + arm; warm-up rollouts
$15 \leq t < 35$ s	this node	100 Hz body-rate + thrust commands from the certified plan; wind EKF and GP learning; replanning
$t \geq 35$ s	PX4	position setpoint above the origin, then land and disarm

At any time in the RTA window, if no certified plan exists for more than 20 ms (`--backup-grace`), the node commands PX4 **LAND** and stops sending commands. 5. **Shutdown** (Ctrl-C, or the end of `launch`): `close()` stops the worker process and prints the timing summary; `save_flight_log()` writes the HDF5 file (and the legacy CSV next to it).

4 The parts and how they connect

Figure 1 (on the next page) shows the node at a glance.

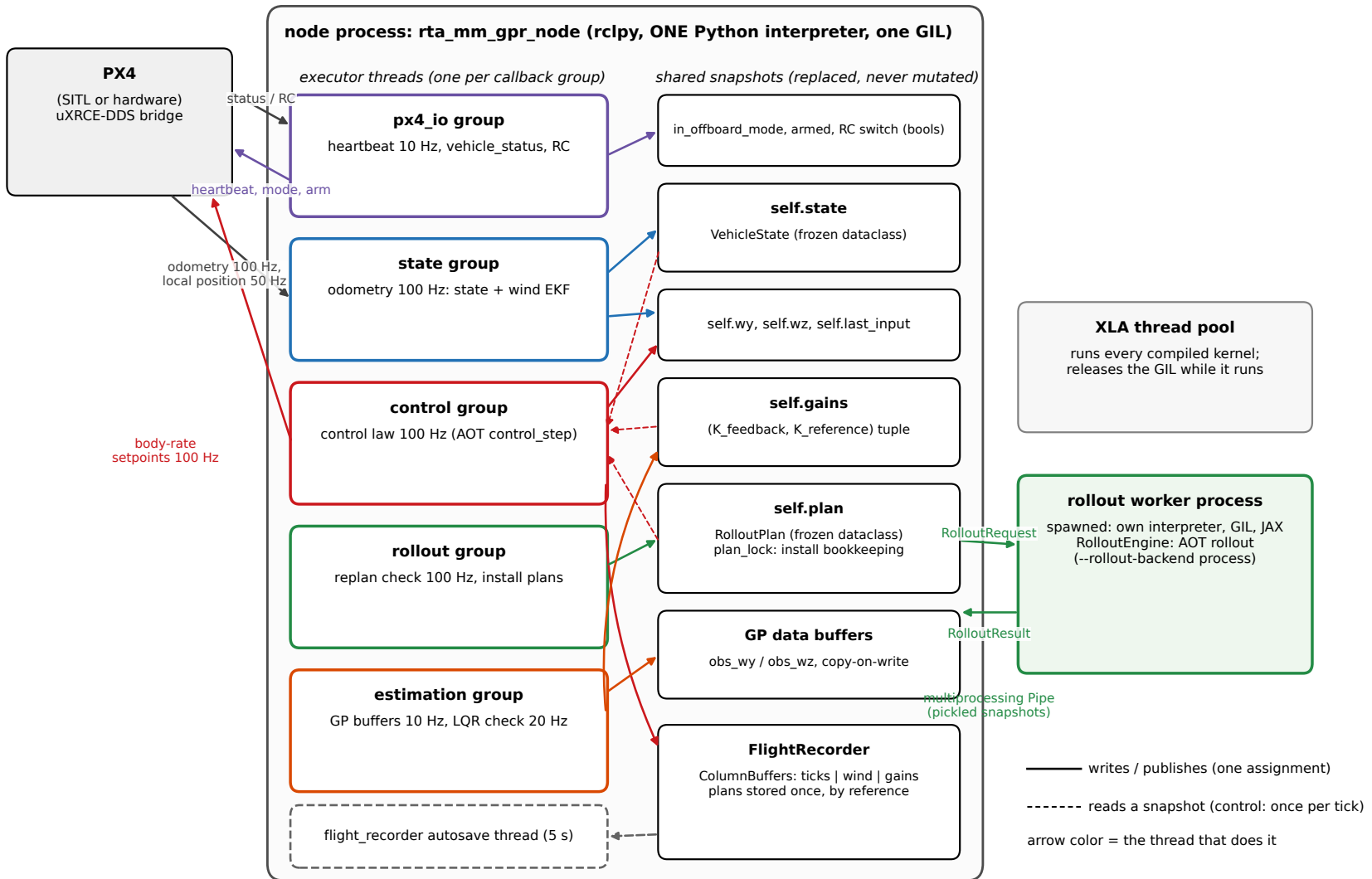


Figure 1: The node process: one executor thread per callback group, the shared snapshots they exchange, the rollout worker process and PX4. Solid arrows: the thread that writes (publishes) a snapshot, in that thread's color. Dashed: the control thread's reads on its hot path. Not drawn, to keep it readable: the rollout thread also reads `self.state`, `self.gains` and the GP buffers when it builds a request; the estimation thread reads `self.state` and the wind estimate.

4.1 One control tick (100 Hz, control group)

`control_algorithm_callback` → `control_administrator` (`scripts/rta_mm_gpr_node.py:989`):

1. read `s = self.state` and `plan = self.plan` **once** (they may be replaced at any moment);
2. if `t > plan.collection_time` the plan is no longer certified: count it, and after 20 ms continuously uncertified, `engage_backup()` → PX4 LAND;
3. `traj_idx = plan.index_at(t)`: the plan row for *now*, computed from time, not from a tick counter;
4. **one** compiled call, `kernels.control_step(...)`: the NR tracker (pitch and yaw rates) and the RTA feedback `u = clip(u_ff[i] - K (x - x_ref[i]))` (thrust and roll rate), clipped;
5. thrust → throttle (`get_throttle_command_from_force`), publish the body-rate setpoint;
6. one row into the recorder's ticks buffer.

4.2 One plan (rollout group, and the worker process)

`rollout_callback` (:765) runs every 10 ms but only starts a rollout when it is due:

1. `poll()` the backend; a finished result is **installed** (`install_plan`, :843): it becomes an immutable `RolloutPlan`, is published with one assignment `self.plan = plan`, and stored in the log;
2. a rollout is due when `t >= collection_time - lead`, where `collection_time` is when the current plan stops being certified and `lead` is about twice the recent worst rollout latency (**pre-emptive replanning**: the next plan is installed before the current one expires);
3. the request is a **snapshot**: `RolloutRequest(t, state, K_feedback, K_reference, obs_wy, obs_wz, goal)`, all values copied or immutable at that moment;
4. the engine runs the compiled rollout: the reference trajectory under the reference LQR, and the **mixed-monotone embedding** that bounds every state the vehicle can reach, given the GP's wind bounds (mean ± 3 plus a Lipschitz buffer). It stops at the **first row that fails the certificate** plus a 1 s margin (early exit);
5. the certificate is **spatial**: a row fails when the tube's position bounds (p_y or p_z) are more than 0.25 m + from the reference position, or the tube reaches the 0.3 m floor. , carried by the request, is max(model-mismatch margin 0.05 m, position-estimate uncertainty: 0 in SITL, 3 of EKF2 on hardware); it also widens the initial box (docs 04, 06).

4.3 Wind and gains (state and estimation groups)

- **Wind EKF** at the odometry rate (100 Hz), in the state thread, only while this node's commands fly: measured acceleration minus the model's (the compiled `wind_model`) is the wind force estimate (`self.wy`, `self.wz`).
- **GP data**, 10 Hz, estimation thread: one row per buffer (time, input, wind): the y-wind is learned as a function of altitude, the z-wind as a function of lateral position. Each buffer is a 9-row ring, updated **copy-on-write** (see Locks).
- **LQR**, estimation thread: re-linearise the planar model (compiled `linearize`) and solve two LQRs (`control.lqr`) every 1.8 s, or when the yaw error exceeds 5°. The pair of gains is published as one tuple.

5 Ahead-of-time JAX compilation

5.1 What JAX does with a function

JAX turns a Python function into an XLA program in two steps:

1. **Tracing.** JAX calls the function with *abstract* arguments (only shapes and dtypes) and records every `jnp` operation into a graph (a *jaxpr*). Python control flow runs once, at trace time; loops that must stay loops use `lax.scan`, `lax.while_loop` or `lax.fori_loop`.
2. **Compilation.** XLA optimises the graph (fusing operations, choosing memory layouts) and emits machine code for the CPU. Running it later costs microseconds to milliseconds; compiling costs 0.1–10 s.

`jax.jit(f)` does both lazily, at the first call, and **caches one executable per call signature**: the shapes and dtypes of the arguments, but also details that are easy to miss: a Python `float` is a *weakly typed* scalar while a `float64` array is not; an array already on the device is *committed*. Any difference means a new trace and a new compile.

5.2 Why the node compiles ahead of time

The original node “warmed up” by calling each jitted function once in `__init__`. In flight, one argument (`last_input`) became the controller’s own output array: a different signature, so XLA compiled again **in the air** (322 ms: 32 missed control ticks), plus more compiles for slicing and concatenation.

AOT compilation removes the possibility:

```
f64 = lambda *shape: jax.ShapeDtypeStruct(shape, jnp.float64)
compiled = jax.jit(control_step).lower(f64(9), f64(5), f64(4),
    ↪ ...).compile()
```

- `lower(...)` traces for exactly these shapes and dtypes, **without calling** the function;
- `.compile()` returns a `jax.stages.Compiled`: one executable, one signature. **It never recompiles.** Called with anything else it raises `TypeError` immediately, which is what you want to find out on the bench, not in flight;
- every in-flight argument goes through `as_f64(x, shape)` (`control_kernels.py:36`): `np.ascontiguousarray(x, dtype=np.float64).reshape(shape)`. One canonical type, always.

5.3 What is compiled

executable	built in	inputs → outputs	used by
<code>control_step</code>	<code>control_kernels.py:54</code>	states, last input, NR reference, plan row, gains → 4 commands	control thread, 100 Hz
<code>wind_model</code>	<code>control_kernels.py:61</code>	NR state, input → model accelerations	state thread, 100 Hz

executable	built in	inputs $\rightarrow$ outputs	used by
linearize	control_kernels.py:65	planar state, input $\rightarrow A, B$ (<code>jax.jacfwd</code>)	estimation thread
rollout	rollout_backends.py:131	t_0 , state, x_{pert} , gains, 2 GP buffers, goal, threshold (y, z) $\rightarrow$ tube, reference, feedforward, violation row, rows used	rollout thread or worker process

`control_step` fuses what used to be about six separate dispatches per tick (NR tracker, slicing, `u_applied`, concatenation, clipping) into one executable: one call from Python per tick.

Static vs traced arguments. In the rollout, the things that never change during a flight (horizon, time step, the planar system, its mixed Jacobian, input limits, threshold, floor) are bound with `functools.partial` *before* lowering: they are constants of the compiled program. Only the eight arrays in the table are traced inputs.

Fixed shapes with early exit. XLA programs have fixed shapes. The early-exit rollout (`jax_mm_rta/mm_rta.py:307`) uses `lax.while_loop` over preallocated full-horizon arrays filled with NaN and returns `n_valid`, the number of rows it actually computed; the engine trims the result in NumPy. The loop stops at the first failing row plus the margin (3–7 ms instead of 60–190 ms for the full 30 s), and the rows it computes are identical to the full rollout’s (it only depends on earlier rows).

5.4 Configuration and the disk cache

`utilities/jax_setup.py` is imported **before anything is compiled** (every module that compiles imports it first):

- `jax_enable_x64 = True`: float64 everywhere (the default is float32);
- `jax_compilation_cache_dir = ~/.cache/px4_rta_mm_gpr/jax`: compiled executables are written to disk and reused by later runs *and by the worker process*. Start-up compilation went from about 20 s to about 4–6 s. `PX4_RTA_JAX_CACHE=''` disables it. Changing the code of a compiled function changes its key, so the cache never serves a stale executable.

5.5 Practical rules

- A new argument shape (e.g. more GP rows, a longer horizon) needs a change where the shapes are declared (`_shapes` in `RolloutEngine`, the `lower(...)` calls in `ControlKernels`). It then fails loudly at start-up, never silently in flight.
- Keep Python control flow on *static* values; anything that depends on array values must be `jnp.where`, `lax.cond` or a `lax` loop.
- JAX arrays are immutable: `a.at[i].set(v)` returns a *new* array. (The original GP buffers used `.at[].set()` and discarded the result, so the GP never learned; they are NumPy arrays now.)

6 Concurrency

6.1 Threads, the GIL and XLA

A Python process runs one interpreter with one **global interpreter lock (GIL)**: only one thread executes Python bytecode at a time. Threads still help here because the heavy work is not Python: a compiled XLA executable **releases the GIL** while it runs, and so do NumPy's large operations and DDS I/O. While the rollout's executable runs on XLA's thread pool, the control thread holds the GIL for its ~0.2 ms tick and continues on schedule. Doc 01 has the full story with timelines.

6.2 Executors

An rclpy **executor** waits for ready timers and messages and runs their callbacks. `--executor:`

option	what it is	callbacks run
multi (default)	<code>MultiThreadedExecutor</code> , 5 threads	in parallel across callback groups
single	<code>SingleThreadedExecutor</code> (the original <code>rclpy.spin</code>)	one at a time
events	<code>EventsExecutor</code> (<code>rclpy.experimental</code>)	one at a time; the wait/dispatch loop is C++ (much less overhead)

6.3 Callback groups

A **mutually exclusive callback group** never runs two of its callbacks at the same time; different groups may run in parallel on a multithreaded executor. Five groups, so five threads are enough:

group	callbacks (period)	writes
px4_io_group	heartbeat (0.1 s), vehicle_status, RC channels	offboard / armed / RC flags; PX4 mode and arm commands
state_group	PX4 vehicle_odometry (100 Hz) + wind EKF; vehicle_local_position (50 Hz, acceleration)	<code>self.state</code> , <code>self.wy</code> , <code>self.wz</code> , <code>self.position_sigma</code> , recorder wind
control_group	control law (0.01 s)	<code>self.last_input</code> , recorder ticks, setpoints to PX4
rollout_group	rollout check (0.01 s)	<code>self.plan</code> , <code>self.collection_time</code> , recorder plans
estimation_group	GP buffers (0.1 s), LQR check (0.05 s)	GP buffers, <code>self.gains</code> , recorder gains

The rule that makes this work: **each piece of shared state has exactly one writer thread**. Readers in other threads only ever take a snapshot.

6.4 Where the rollout runs (`--rollout-backend`)

`concurrency/rollout_backends.py` hides *where* the rollout runs behind `submit(request) / poll()`:

- **thread**: `submit` computes synchronously in the rollout group’s executor thread. With `multi`, that thread is separate from control, and XLA releases the GIL, so control keeps its schedule.
- **process** (recommended, with `--executor events`): a separate OS process with its own interpreter, GIL and JAX runtime. `submit` sends the pickled `RolloutRequest` through a `multiprocessing.Pipe` and returns at once; `poll()` checks the pipe without blocking. Even a single-threaded executor then never waits for a rollout. Details:
 - start method **spawn**, not **fork**: forking a process that already runs JAX/XLA and DDS threads can deadlock the child; **spawn** starts a fresh interpreter that imports only what it needs;
 - the worker is started **first** in `__init__`, so its compilation overlaps with the node’s;
 - `RolloutConfig` (static) is sent once at start; each request/result is a small frozen dataclass of NumPy arrays (cheap to pickle); the result arrays are trimmed to `n_valid` rows;
 - `--rollout-cpus 2,3` pins the worker to CPUs (`os.sched_setaffinity`), away from the control thread;
 - a worker exception is sent back as `('error', traceback)` and raised in the node, never lost.

6.5 Garbage collection

CPython frees most objects by reference counting; a separate **cycle collector** runs in generations. A full (generation 2) collection walks every tracked object **while holding the GIL**: measured ~300 ms, i.e. 30 missed ticks, and the JAX/immrax start-up creates hundreds of thousands of long-lived objects.

- `gc.freeze()` after start-up moves all of them to a permanent generation that is never scanned again (about 296 000 objects);
- `--gc-no-full` raises the generation-2 threshold so no full collection runs in flight (young collections still run, ~1 ms); one full collection runs at shutdown;
- a `gc.callbacks` hook measures every in-flight pause and puts it in the log.

6.6 Timing statistics

`LoopStats` (`concurrency/loop_stats.py`) wraps each periodic callback in a context manager that records the actual period and execution time into preallocated arrays, under its own small lock. The summary at shutdown (“control: rate 100.0 Hz, period p99 10.3 ms, misses 0”) and the log’s `records/timing/` come from it.

7 Shared state and locks

7.1 The pattern: build, then publish with one assignment

Most of the node's sharing uses **no lock** at all. A writer builds a complete, immutable object and publishes it by assigning one attribute; a reader reads the attribute **once** and works with that object:

```
# rollout thread (writer)                # control thread (reader)
plan = RolloutPlan(...) # complete      plan = self.plan          # ONE read
self.plan = plan           # publish     row = plan.index_at(t)  # consistent
```

In CPython, assigning an attribute (a reference) is atomic with respect to other threads: a reader sees the old object or the new one, never a mixture. The objects are never modified after publication (frozen dataclasses, tuples; by convention the NumPy arrays inside them are not written either), so the reader needs no lock for as long as it uses them. The same idea in C++ is `std::atomic<std::shared_ptr<const T>>` (the fast loop on cpp-version).

What goes wrong without it: the original node kept `self.x`, `self.y`, `self.vx`, ... as separate attributes updated one by one; another thread could read a new position with an old velocity. The same for a plan whose tube and reference were assigned separately.

7.2 Every shared variable

shared	written by	read by	mechanism
<code>self.state</code> (<code>VehicleState</code>)	state	control, rollout, estimation	frozen dataclass, one assignment
<code>self.plan</code> (<code>RolloutPlan</code>)	rollout	control, <code>px4_io</code> (status line)	frozen dataclass, one assignment; <code>plan_lock</code> around installation
<code>self.gains</code> ((<code>K_fb</code> , <code>K_ref</code>))	estimation	control, rollout	tuple, one assignment (both gains always from the same linearisation)
GP buffers <code>obs_wy</code> , <code>obs_wz</code>	estimation	rollout (into the request)	copy-on-write: copy, modify the copy, assign
<code>self.wy</code> , <code>self.wz</code>	state	estimation, control (logged)	floats, one assignment each
<code>self.position_sigma</code> (<code>EKF2</code> , <code>y</code> and <code>z</code>)	state	rollout (into the request's)	a new array, one assignment
<code>self.last_input</code>	control	state (wind model)	a new array each tick, one assignment
offboard / armed / RC flags	<code>px4_io</code>	control, heartbeat	bools
warm-up request	estimation	rollout	<code>threading.Event</code>

shared	written by	read by	mechanism
recorder streams	one thread each (ticks: control, wind: state, gains: estimation)	autosave thread	ColumnBuffer : row written, <i>then</i> the count published
recorder records / events	rollout (plans), control (events)	autosave thread	Recorder._lock
LoopStats samples	the measured callback	shutdown summary	LoopStats._lock

7.3 The locks there are, and why

- **plan_lock** (`threading.Lock`, `rta_mm_gpr_node.py:324`): held while a plan is installed, so the plan number, `collection_time` and the new plan are updated together. The control thread never takes it; it reads `self.plan` once (the plan itself carries its own `collection_time`).
- **Recorder._lock** (`flight_recorder`): guards the lists of records and events and the stream registry. Appending a record and the autosave thread’s snapshot of “what is new since the last flush” are short critical sections.
- **Recorder._flush_lock**: one flush at a time (the autosave thread and the final `save()` could otherwise both write the file).
- **Recorder._stop** (`threading.Event`): tells the autosave thread to stop; `save()` joins it before the final flush.
- **LoopStats._lock**: the measured thread appends while the shutdown code may read.
- **No lock in ColumnBuffer.append**: each stream has one writer; the row is written first and the row count incremented after it, so a concurrent reader (the autosave flush) sees only complete rows. When the buffer grows, it copies into a new array and swaps the reference; a reader holding the old array still has valid rows.

A known, harmless race. `self.wy`, `self.wz = wy`, `wz` is two assignments, so a reader could pair a new `wy` with the previous `wz` (25 ms older). Both are slowly varying estimates that only feed the GP buffers and the log; if they ever had to be consistent, they would become one tuple, like the gains.

8 The control law, the model and the GP (in brief)

- **Planar model** (`jax_mm_rta/mm_rta.py:75`): state (p_y, p_z, h, v, θ) with body-frame velocities h, v ; inputs thrust u_1 and roll rate u_2 ; wind forces w_y, w_z .
- **TVGPR** (`jax_mm_rta/TVGPR.py`): time-varying GP regression, squared-exponential kernel in space times a forgetting factor $(1 - \epsilon)^{|t-t'|/2}$ in time, 9 observations per GP. `GPR.py` is the time-invariant variant, kept as part of the package API.
- **Rollout step** (`_make_step`): the reference moves under the reference LQR (plus the GP mean feedforward on thrust); the tube moves under one of the paper’s **embedding systems** (Appendix A), built from `immrax`’s mixed Jacobians (`irx.mjacM`) of the dynamics: (66)-(67) by default (first order in the feedback, the GP disturbance bounded over the box), or (68)-(69)

/ (64)-(65) with `--embedding`. With TV-GPR they are equally safe and certify the same horizon; (66)-(67) is the cheapest (doc 06).

- **NR tracker** (`jax_nr/`): predicts the output 0.8 s ahead (forward Euler in a `lax.fori_loop`), inverts the Jacobian of that prediction with respect to the input, and adds an integral control-barrier term that keeps the rates within ± 0.8 rad/s. It flies pitch and yaw; the RTA feedback flies thrust and roll.

9 Logging, analysis and simulation reuse the same parts

- **Flight log** (`flight_log/recorder.py` on `flight_recorder`): streams ticks, wind, gains; every plan stored once with the inputs that produced it (`records/plans/<seq>`); timing arrays; events; all options as metadata. Autosaved every 5 s.
- **Numerical simulation** (`sim/planar.py`): the same `RolloutEngine`, `ControlKernels`, `u_applied` and `FlightRecorder`, around an RK4 integration of the planar model and a known wind field (`sim/wind.py`). It writes the same log format, so the analysis works on it unchanged.
- **Analysis** (`analysis.py`): summaries, plots, the `PlanGP` reconstruction of the exact GP a plan used, `tube_violation` (why a certificate ended) and `animate` (the tube videos).

10 Every import, and what it is for

10.1 Standard library

import	what it is	used for
<code>os</code>	operating-system interface	paths, <code>makedirs</code> , environment (<code>PX4_RTA_JAX_CACHE</code>), <code>sched_setaffinity</code> (worker CPUs)
<code>sys</code>	interpreter access	<code>sys.argv</code> for the command line
<code>time</code>	clocks	<code>time.time()</code> (node clock, latencies), <code>perf_counter()</code> (compute times, compile times)
<code>threading</code>	threads and synchronisation	<code>Lock</code> (<code>plan_lock</code> , <code>LoopStats</code>), <code>Event</code> (warm-up request), recorder autosave thread
<code>multiprocessing</code>	processes	<code>get_context('spawn')</code> , <code>Process</code> , <code>Pipe</code> for the rollout worker

import	what it is	used for
<code>gc</code>	the cycle garbage collector	<code>freeze</code> , <code>set_threshold</code> , <code>callbacks</code> (pause measurement), <code>get_objects</code> (heap summary)
<code>argparse</code>	command-line parsing	every <code>--option</code> of the node
<code>dataclasses</code>	<code>@dataclass</code>	frozen, immutable snapshots: <code>RuntimeOptions</code> , <code>VehicleState</code> , <code>RolloutPlan</code> , <code>RolloutConfig/Request/Result</code> , <code>SimConfig</code>
<code>typing</code>	type hints	<code>Optional</code> , <code>Sequence</code> , <code>Dict</code> , <code>Callable</code> , <code>Tuple</code> (documentation and editors)
<code>from __future__ import annotations</code>	lazy annotations	type hints are not evaluated at import time (cheaper, allows forward references)
<code>partial</code> (<code>functools</code>)	bind arguments	fix the rollout's static arguments before lowering; <code>@partial(jax.jit, static_argnames=...)</code>
<code>contextmanager</code> (<code>contextlib</code>)	generator-based <code>with</code> blocks	<code>LoopStats.measure()</code>
<code>inspect</code> , <code>traceback</code>	introspection, stack traces	name the failing function and print the full trace in error handlers; ship worker errors to the node
<code>math</code>	scalar math	thrust $\rightarrow$ throttle conversion in the vehicle constants
<code>glob</code>	file patterns	<code>setup.py</code> (launch files), notebooks (log files)
<code>textwrap</code> <code>Counter</code> (<code>collections</code>)	text wrapping counting	reasons in the videos heap summary (object types the GC scans)

10.2 ROS 2

import	what it is	used for
<code>rclpy</code>	the ROS 2 Python client library	<code>init</code> , <code>ok</code> , <code>shutdown</code>
<code>Node</code> (<code>rclpy.node</code>)	base class of a node	<code>OffboardControl</code> subclasses it: publishers, subscriptions, timers, logger

import	what it is	used for
<code>MutuallyExclusiveCallbackGroup</code> (<code>rclpy.callback_groups</code>)	callback groups	the five groups above
<code>QoSProfile</code> , <code>ReliabilityPolicy</code> , <code>HistoryPolicy</code> , <code>DurabilityPolicy</code> (<code>rclpy.qos</code>)	quality of service	PX4's topics need best-effort, transient-local, depth 1
<code>SingleThreadedExecutor</code> , <code>MultiThreadedExecutor</code> (<code>rclpy.executors</code>); <code>EventsExecutor</code> (<code>rclpy.experimental</code>)	executors	<code>--executor</code>
<code>px4_msgs.msg</code>	PX4's uORB messages as ROS 2 types	<code>OffboardControlMode</code> , <code>VehicleCommand</code> , <code>TrajectorySetpoint</code> , <code>VehicleRatesSetpoint</code> (out); <code>VehicleStatus</code> , <code>RcChannels</code> , <code>VehicleOdometry</code> (state + EKF2 position variance, 100 Hz), <code>VehicleLocalPosition</code> (acceleration, 50 Hz) (in)
<code>launch</code> , <code>Node</code> (<code>launch_ros.actions</code>)	ROS 2 launch system	the launch files
<code>setuptools</code>	Python packaging	<code>setup.py</code> (<code>find_packages</code> , entry points, data files)

10.3 Numerics and control

import	what it is	used for
<code>numpy</code>	arrays	everything outside compiled code: states, buffers, the EKF, logging, canonical <code>float64</code> arguments
<code>jax</code> , <code>jax.numpy</code> as <code>jnp</code>	NumPy-like arrays + tracing + XLA compilation + automatic differentiation	the model, GPs, rollout, NR tracker; <code>jax.jit(...).lower(...).compile()</code> , <code>ShapeDtypeStruct</code> , <code>jacfwd</code> , <code>vmap</code> , <code>config</code>
<code>jax.lax</code> (<code>scan</code> , <code>while_loop</code> , <code>fori_loop</code> , <code>cond</code>)	structured control flow that stays inside the compiled program	the full-horizon rollout, the early-exit rollout, NR's prediction loop, the certificate test
<code>jacfwd</code> , <code>jacrev</code> , <code>hessian</code> (<code>jax</code>)	automatic differentiation	NR's Jacobian of the prediction; <code>linearize</code> for the LQR

import	what it is	used for
<code>immrax as irx</code>	interval analysis and mixed-monotone reachability on JAX	<code>irx.interval</code> , <code>icentpert</code> (initial box), <code>i2ut/ut2i</code> (interval embedding vector), <code>mjacM/jacM</code> (mixed Jacobian bounds), <code>ifemb</code> (embedding system), <code>PermutationSystem</code> (base class of the planar model)
<code>control</code> (python-control)	control-systems toolbox	<code>control.lqr(A, B, Q, R)</code> for both gain matrices
<code>Rotation</code> (<code>scipy.spatial.transform</code>)	rotations	quaternion $\rightarrow$ roll, pitch, yaw in the odometry callback

10.4 Data, plots and video

import	what it is	used for
<code>flight_recorder: Recorder</code> , <code>ColumnBuffer</code> , <code>git_commit</code>	this project's logging library (submodule)	preallocated streams, records, events, HDF5 flushing, autosave
<code>h5py</code> (inside <code>flight_recorder</code> and the reader)	HDF5 files	the flight log
<code>pandas</code>	data frames	log tables (<code>log.ticks</code>), summaries
<code>matplotlib</code> (<code>pyplot</code> , <code>colors</code> , <code>PolyCollection</code> , <code>patches</code> , <code>lines</code>)	plotting	figures and video frames (tube boxes are <code>PolyCollections</code>)
<code>imageio_ffmpeg</code>	a bundled ffmpeg binary + a frame writer	MP4 videos (<code>write_frames</code>)
<code>PIL</code> (Pillow)	images	GIF previews (resize, 96-colour palette)

10.5 This package's own modules (who imports what)

module	imports from the package	imported by
<code>utilities/jax_setup.py</code>	–	everything that compiles (imported first)
<code>jax_mm_rta/</code>	<code>TVGPR</code> , <code>jax_setup.jit</code>	<code>node</code> , <code>control_kernels</code> , <code>RolloutEngine</code> , <code>sim</code> , <code>tests</code>
<code>jax_nr/</code>	–	<code>control_kernels</code> , <code>node</code> (dynamics)
<code>control_kernels.py</code>	<code>jax_setup</code> , <code>jax_nr</code> , <code>jax_mm_rta.u_applied</code>	<code>node</code> , <code>sim</code>

module	imports from the package	imported by
concurrency/	jax_mm_rta (inside RolloutEngine, so a spawned worker imports JAX itself)	node, sim
flight_log/	flight_recorder	node, sim, analysis
px4_functions/	px4_msgs	node only (needs ROS)
utilities/	–	node (adjust_yaw, vehicle constants)
sim/	concurrency, control_kernels, flight_log, jax_mm_rta	notebook 04, tests
analysis.py	flight_log	notebooks, tests

from px4_rta_mm_gpr.jax_mm_rta import * and from px4_rta_mm_gpr.px4_functions import * in the node are the two star imports left from the original code; `__all__` in each `__init__.py` lists what they bring in.

11 On cpp-version: the C++ fast loop

This branch is `main` plus one layer: the time-critical half of the node moves to C++, and the Python node becomes the **planner**. Doc 05 has the design and the measurements; this section shows how it fits into the structure above.

Two more packages (next to the others in `packages/`):

```
px4_rta_mm_gpr_msgs/  PlannerStatus, RtaPlan, RtaGains, ControlTick
px4_rta_mm_gpr_cpp/   rta_fast_loop, control_law.hpp, control_law_check
```

Both are `ament_cmake` packages.

Two processes. `ros2 launch px4_rta_mm_gpr cpp_rta_launch.py` starts `rta_fast_loop` and `px4_rta_mm_gpr --cpp-control`:

process	owns
<code>rta_fast_loop</code> (C++)	everything PX4 sees: heartbeat, offboard/arm, odometry, the 100 Hz control law (NR tracker with an exact dual-number Jacobian + RTA feedback), the certification watchdog and LAND backup
<code>px4_rta_mm_gpr --cpp-control</code> (Python)	the mission clock and every tunable, rollouts (thread or worker process), wind EKF and GP data, LQR gains, the flight log

Topics between them (reliable QoS):

topic	message	direction	rate
/rta/planner_status	PlannerStatus: mission t_0 , mass, limits, NR parameters, phase times, backup settings	planner $\rightarrow$ fast loop	10 Hz; the first one configures the fast loop
/rta/plan	RtaPlan: seq, t_0 , dt, collection_time, violation row, reference and feedforward rows	planner $\rightarrow$ fast loop	every installed plan
/rta/gains	RtaGains: the 2×5 feedback gain	planner $\rightarrow$ fast loop	every LQR update
/rta/control_tick	ControlTick: state, command, plan row, timing, phase, backup	fast loop $\rightarrow$ planner	100 Hz; logged into the same HDF5 file

Threads and snapshots in C++. `rta_fast_loop` uses a `MultiThreadedExecutor` with two mutually exclusive groups: subscriptions in one, the control and heartbeat timers in the other, so deserialising a plan never delays a tick. It uses the same build-then-publish pattern as the Python node, with a type that makes it explicit:

```
std::atomic<std::shared_ptr<const Plan>> plan_;    // C++20

// subscription thread (writer)
auto p = std::make_shared<Plan>(/* complete */);
plan_.store(p);                                   // publish

// control thread (reader)
const auto plan = plan_.load();                   // ONE snapshot
const auto row = plan->ref_row(plan->index_at(t));
```

`shared_ptr` keeps a plan alive while any thread still uses it (no reader can see it freed), and `const` makes it immutable. The flags (`nav_state_`, `armed_`, `rc_offboard_`) are `std::atomic` scalars. One ordering detail: on the first `PlannerStatus`, the derived constants (limits, NR parameters) are written *before* `config_.store(msg)`; the control timer does nothing until it loads a non-null `config_`, and the atomic store/load pair guarantees it then sees those constants.

The planner side (`rta_mm_gpr_node.py` with `--cpp-control`): the PX4 publishers and the control and heartbeat timers are not created. Instead: a 10 Hz `planner_status_callback` (`px4_io` group), `publish_plan` in `install_plan` and `publish_gains` in the LQR update, and `control_tick_callback` (control group), which logs each tick and feeds `LoopStats.record()` with the fast loop's measured period and compute time. Rollouts, AOT compilation, the executors and every lock described above are unchanged.

C++ includes (`rta_fast_loop.cpp`, `control_law.hpp`):

include	what it is	used for
<code>rclcpp/rclcpp.hpp</code>	the ROS 2 C++ client library	node, publishers, subscriptions, wall timers, callback groups, <code>MultiThreadedExecutor</code>
<code>px4_msgs/msg/*.hpp</code>	PX4 message types	offboard mode, vehicle command, setpoints, vehicle status, RC
<code>px4_msgs/msg/vehicle_odometry.hpp</code>	PX4 odometry (100 Hz)	the fast loop's state
<code>px4_rta_mm_gpr_msgs/msg/*.hpp</code>	this project's messages	the four topics above
<code>Eigen/Dense</code>	linear-algebra templates	fixed-size vectors and matrices in the control law
<code><atomic></code> , <code><memory></code>	atomics, smart pointers	<code>std::atomic<std::shared_ptr<const T>></code> snapshots
<code><chrono></code>	clocks	mission clock (system clock, as the planner), compute time (steady clock)
<code><optional></code>	a value or nothing	<code>uncertified_since_</code> (the watchdog's start time)
<code><cmath></code> , <code><limits></code> , <code><array></code> , <code><algorithm></code>	math, infinities, fixed arrays, <code>clamp</code>	the control law and limits

`control_law.hpp` also defines `Jet<N>`, a dual number (value + gradient): running the NR predictor on `Jet<4>` values gives the prediction and its exact Jacobian with respect to the 4 inputs in one pass, the C++ counterpart of `jax.jacfwd`. `control_law_check + test/test_control_law.py` compare it with the JAX kernels on 500 random cases.

12 Where to change what

to ...	change
add a command-line option	<code>argparse</code> in <code>scripts/px4_rta_mm_gpr.py</code> , a field in <code>RuntimeOptions</code> , use <code>self.options.<name></code>
compare wind models / embedding systems	<code>--gp tv\ static</code> , <code>--embedding uw\ u\ none</code> (node); <code>SimConfig(gp_epsilon=..., embedding=...)</code> ; notebook 05
change what the certificate tests	<code>_row_fails</code> and <code>collection_id_jax</code> (<code>jax_mm_rta/mm_rta.py</code>), <code>tube_violation</code> (<code>analysis.py</code>), the test in <code>test/test_sim_analysis.py</code>
log a new per-tick value	<code>TICK_COLUMNS</code> and <code>tick()</code> in <code>flight_log/recorder.py</code> (same order)
add a new periodic job	a timer in a callback group; if it writes shared state, publish it as one immutable object

to ...	change
change an input shape of a compiled kernel	the <code>lower(...)</code> shapes (<code>ControlKernels</code> , <code>RolloutEngine._shapes</code>) and the <code>as_f64</code> reshapes
run without ROS	<code>px4_rta_mm_gpr.sim.simulate(SimConfig(...), winds=...)</code>

13 Glossary

AOT ahead-of-time compilation: compiling for a declared signature before the first call.

callback group a set of callbacks that an executor never runs concurrently (mutually exclusive).

certificate / certified horizon the time span over which the reachable tube stays within 0.25 m of the reference position and above the floor; the plan may be flown only within it.

copy-on-write modify a copy, then replace the shared reference with it.

embedding system a system of twice the dimension whose state is a lower and an upper bound; its trajectory bounds every trajectory of the original system (mixed monotonicity).

GIL CPython’s global interpreter lock: one thread runs Python bytecode at a time.

snapshot an immutable object read once and used for the rest of a callback.

XLA the compiler behind JAX; compiled programs run on its own threads and release the GIL.